A BOOST FOR THE FINAL ALL-OUT EFFORTS!

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When exercising above the first—and especially the second—lactate threshold (or above FTP/critical power), there is an exponential rise in hydrogen ions (H^+), which increases acidity and contributes to muscle fatigue.

As highlighted in a previous KIW Research Note by Dr Jeff Rothschild, **sodium bicarbonate acts as an acid buffer**. Ingesting it raises blood bicarbonate levels, which reduces H^+ concentration in the blood and enhances the exchange of hydrogen ions between muscle and blood. **This helps delay fatigue during efforts above the second threshold or FTP/CP.**

Typically, the greatest benefit of sodium bicarbonate occurs for efforts lasting roughly 30 seconds to 12 minutes. Longer efforts generate less acidosis and are therefore less sensitive to improvement via bicarbonate supplementation. However, mass-start road cycling races present a unique scenario: they last several hours, the average intensity is below the first threshold but they included many efforts at medium and high intensity especially at the very end of the race.

Peak blood bicarbonate levels typically occur 60–150 minutes after ingestion, with considerable individual variability. For this reason, simply taking bicarbonate before a long mass-start road race may not be the most effective strategy to support the decisive

high-intensity effort at the finish. This raises a key question:

Does ingesting bicarbonate during a long race improve performance in the final all-out effort/s?

A study published by Dalle and Colleagues (Department of Movement Sciences, KU Leuven, Belgium) in 2021 on *Journal of Science and Medicine in Sport* tried to answer this question. (1)

WHAT DID THEY DO?

- They recruited eleven male cyclists (age: 22, VO2max: 64)
- Each participant completed **two simulated 3-hour races** on separate days, with exercise intensity varying every 5 minutes between 60% and 90% of the first lactate threshold. At the end of each 3-hour session, a **90-second all-out test** was performed to evaluate performance at the finish of a long race.

  The difference between the two days was in supplementation:

  - **BICARB Day**

    Before the race: Sodium bicarbonate was ingested at a total dose of **150 mg/kg body mass**, split as follows:

    - 90 mg/kg with a high-carbohydrate breakfast **2 hours before exercise**
    - Two additional doses of 30 mg/kg at 45 and 90 minutes after breakfast

    During the race: 50 mg/kg body mass per hour

  - **CONTROL (CON) Day**

    Participants received a **placebo** identical in appearance to the bicarbonate at the same timepoints.

  On both days, cyclists also ingested **60 g of carbohydrates per hour** during the trial.

WHAT DID THEY FIND?

In line with the supplementation strategy used, **at the start of 90-sec bout at the end of the 3h simulated race, blood bicarbonate concentration and pH were both higher in BICARB than in CONTROL:**

- blood bicarbonate concentration: 31.5 vs 24.4 mmol/L
- blood pH: 7.50 vs 7.41

- **Mean power output during the 90-sec bout at the end of the simulated races was ~3% higher in BICARB VS CONTROL with 8 out of 11 participants who improved their performance with the supplementation strategy:**

    541 ± 59 W vs 524 ± 57 W; +17 ± 25 W

- In line with this, peak heart rate during the 90-sec bout at the end of the simulated races was higher in BICARB vs CONTROL:

    184 ± 7 bpm vs 181 ± 5 bpm; +3.1 ± 2.6 bpm;

PRACTICAL TAKEAWAYS

Top-up bicarbonate supplementation at 50 mg/kg per hour during long (>3 h) races can be an effective strategy to boost all-out performance in the final stages of the race.

GABRIELE'S COMMENT

No much more to add. I think these data clearly suggest that bicarbonate isn't just for short time trials. Taken at breakfast **and during** a long race, it can help cyclists deliver stronger efforts at high intensity after hours of racing.

Thanks for Reading! Share with your Team!

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Founder of Knowledgeiswatt

REFERENCE

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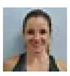
1. Dalle S, Koppo K, Hespel P. Sodium bicarbonate improves sprint performance in endurance cycling. J Sci Med Sport. 2021 Mar;24(3):301-306.



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